



THE TANGENT LINE AND THE DERIVATIVE

Basic Calculus

THE TANGENT LINE AND THE DERIVATIVE

- In junior high school math, you have learned that one way to describe a line is through its slope. The slope is a numerical expression of a line's steepness. It is given by a constant value, m , taken from the quotient of the y -difference and the x -difference of any two points on a line. Curves, on the other hand, do not have slopes. We can only determine the slope of the line that is tangent to the curve at any given point.

THE TANGENT LINE AND THE DERIVATIVE

- This module explains how the limit process is applied to calculate the slope of a line tangent to a point, which leads to the generalization of the derivative.

SLOPE OF THE LINE TANGENT TO A CURVE

- The slope m of the line tangent to the curve $y = f(x)$ at point $P[x, f(x)]$ is given by the equation

$$m = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

provided that the limit exists. This is commonly called the limit of the difference quotient.

SLOPE OF THE LINE TANGENT TO A CURVE

- The slope of the tangent line to a curve at P is also used to find the slope of the curve at that point. Consider the following illustrative examples applying this formula.

EXAMPLE 1:

- Consider the curve of the function $f(x) = 2x^2$.
 - a. Find an expression for the slope of the curve at any point x .
 - b. Identify the slope at the point where $x = 2$.

SOLUTION:

• $f(x) = 2x^2$, so $f(x + h) = 2(x + h)^2$; thus,

$$\begin{aligned} m &= \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{2(x + h)^2 - 2x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{2(x^2 + 2xh + h^2) - 2x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{2x^2 + 4xh + 2h^2 - 2x^2}{h} \end{aligned}$$

SOLUTION:

- $$\begin{aligned} &= \lim_{h \rightarrow 0} \frac{4xh + 2h^2}{2h(2x + h)} \\ &= \lim_{h \rightarrow 0} \frac{h}{h} \cdot \frac{2x + h}{2x + h} \\ &= \lim_{h \rightarrow 0} 2(2x + h) \\ &= 2(2x + 0) \\ &= 4x \end{aligned}$$
- The general expression for the slope of the line tangent to the curve $f(x) = 2x^2$ is $4x$.

SOLUTION:

- To get the slope of the tangent line at a specific point simply use the substitution. Therefore, the slope at $x = 2$ is $4(2) = 8$.

EXAMPLE 2:

- Find the slope of a tangent line to the curve $f(x) = \frac{2}{x}$ at $x = -1$.

SOLUTION:

• $f(x) = \frac{2}{x}$, so $f(x + h) = \frac{2}{x+h}$. Also,

$$\begin{aligned} f(x + h) - f(x) &= \frac{2}{x+h} - \frac{2}{x} \\ &= \frac{2x - 2(x+h)}{x(x+h)} \\ &= \frac{2x - 2x - 2h}{x(x+h)} \\ &= \frac{-2h}{x(x+h)} \end{aligned}$$

SOLUTION:

$$\begin{aligned}
 m &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \left[\frac{\frac{-2h}{x(x+h)}}{h} \right] \\
 &= \lim_{h \rightarrow 0} \left[\frac{1}{h} \cdot \frac{-2h}{x(x+h)} \right] \\
 &= \lim_{h \rightarrow 0} \frac{-2}{x(x+h)} = \frac{-2}{x(x+0)} = -\frac{2}{x^2}
 \end{aligned}$$

- Thus, the slope of the tangent line at $x = -1$ is $m = -\frac{2}{x^2} = -\frac{2}{(-1)^2} = -2$.

SLOPE OF THE LINE TANGENT TO A CURVE

- The equation $m = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ is sometimes very time-consuming because of the presence of the increment h . A more manageable alternate formula can be derived if $z = x + h$. This implies that $h = z - x$. Evaluating the limit as z approaches x , the formula for the difference quotient can now be written as

$$m = \lim_{z \rightarrow x} \frac{f(z) - f(x)}{z - x}$$

EXAMPLE 3:

- Use the alternative formula to find the expression for the slope of the tangent line to the curve $f(x) = \frac{2}{x}$.

SOLUTION:

$$\begin{aligned}
 m &= \lim_{z \rightarrow x} \frac{f(z) - f(x)}{z - x} = \lim_{z \rightarrow x} \frac{\frac{2}{z} - \frac{2}{x}}{z - x} \\
 &= \lim_{z \rightarrow x} \left[\frac{\frac{1}{z} - \frac{1}{x}}{z - x} \cdot \frac{2x - 2z}{1} \right] \\
 &= \lim_{z \rightarrow x} \left[\frac{\frac{1}{z} - \frac{1}{x}}{z - x} \cdot \frac{-2(x - z)}{1} \right] \\
 &= \lim_{z \rightarrow x} \left[\frac{-2}{xz} \right]
 \end{aligned}$$

SOLUTION:

- Because the limit is evaluated as z approaches x , you can substitute z for x . Thus,

$$m = -\frac{2}{x \cdot x} = -\frac{2}{x^2}$$

- The answer is the same as the previous example, with fewer number of steps.

EXAMPLE 4:

- A function is defined by the equation $y = \sqrt{x}$. Where does the slope of the tangent line equal to $\frac{1}{2}$.

SOLUTION:

- First, find the general expression for the slope of the tangent line to the curve at any point using alternative formula.

$$m = \lim_{z \rightarrow x} \frac{f(z) - f(x)}{z - x} = \lim_{z \rightarrow x} \left[\frac{\sqrt{z} - \sqrt{x}}{z - x} \right]$$

SOLUTION:

- Applying the rationalization technique of evaluating limit, you have

$$\begin{aligned}
 \lim_{z \rightarrow x} \left[\frac{\sqrt{z} - \sqrt{x}}{z - x} \right] &= \lim_{z \rightarrow x} \left[\frac{\sqrt{z} - \sqrt{x}}{z - x} \cdot \frac{\sqrt{z} + \sqrt{x}}{\sqrt{z} + \sqrt{x}} \right] \\
 &= \lim_{z \rightarrow x} \left[\frac{1}{(\sqrt{z} + \sqrt{x})} \right] \\
 \lim_{z \rightarrow x} \left[\frac{1}{(\sqrt{z} + \sqrt{x})} \right] &= \frac{1}{(\sqrt{x} + \sqrt{x})} = \frac{1}{2\sqrt{x}}
 \end{aligned}$$

SOLUTION:

- Thus, the general expression for the slope of the tangent line to the curve is $\frac{1}{2\sqrt{x}}$. To answer the point where the slope of the tangent line equals $\frac{1}{2}$, get the value of x such that $\frac{1}{2\sqrt{x}} = \frac{1}{2}$. By inspection, $x = 1$. The slope of the tangent line will be $\frac{1}{2}$ at the point on the curve where $x = 1$. This means that the point of tangency where the slope is $\frac{1}{2}$ is $(1, 1)$.

EXAMPLE 5:

- A function is defined by the equation $y = \sqrt{3x + 4}$. Where does the slope of the tangent line equal to 1?

SOLUTION:

$$\begin{aligned}
 \bullet \quad m &= \lim_{z \rightarrow x} \frac{f(z) - f(x)}{\frac{z - x}{\sqrt{3z + 4} - \sqrt{3x + 4}}} \\
 &= \lim_{z \rightarrow x} \frac{z - x}{\sqrt{3z + 4} - \sqrt{3x + 4}} \cdot \frac{\sqrt{3z + 4} + \sqrt{3x + 4}}{\sqrt{3z + 4} + \sqrt{3x + 4}} \\
 &= \lim_{z \rightarrow x} \frac{z - x}{(3z + 4) - (3x + 4)} \cdot \frac{\sqrt{3z + 4} + \sqrt{3x + 4}}{\sqrt{3z + 4} + \sqrt{3x + 4}} \\
 &= \lim_{z \rightarrow x} \frac{z - x}{(z - x)(\sqrt{3z + 4} + \sqrt{3x + 4})}
 \end{aligned}$$

SOLUTION:

$$\begin{aligned}
 & \bullet \quad = \lim_{z \rightarrow x} \frac{3z - 3x}{(z - x)(\sqrt{\frac{3z + 4}{3}} + \sqrt{\frac{3x + 4}{3}})} \\
 & = \lim_{z \rightarrow x} \frac{3}{(\sqrt{\frac{3z + 4}{3}} + \sqrt{\frac{3x + 4}{3}})} \\
 & = \frac{3}{(\sqrt{\frac{3x + 4}{3}} + \sqrt{\frac{3x + 4}{3}})} \\
 m & = \frac{3}{2\sqrt{\frac{3x + 4}{3}}}
 \end{aligned}$$

SOLUTION:

- Now, to get the value x where the slope is equal to 1, you have

$$\frac{3}{2\sqrt{3x+4}} = 1$$

$$\sqrt{3x+4} = \frac{3}{2}$$

$$3x+4 = \frac{9}{4}$$

$$3x = -\frac{4}{7}$$

$$x = -\frac{4}{12}$$

SOLUTION:

- Thus, the slope of the tangent line to the curve $y = \sqrt{3x + 4}$ is 1 at the point where $x = -\frac{7}{12}$.

EXAMPLE 6:

- Find the equation of the tangent line to the curve $y = \frac{5x-1}{x+2}$ at the point where $x = 9$.

SOLUTION:

$$\begin{aligned}
 \bullet \quad m &= \lim_{z \rightarrow x} \frac{f(z) - f(x)}{z - x} \\
 &= \lim_{z \rightarrow x} \frac{1}{z - x} \left[\frac{5z - 1}{z + 2} - \frac{5x - 1}{x + 2} \right] \\
 &= \lim_{z \rightarrow x} \frac{1}{z - x} \left[\frac{(5z - 1)(x + 2) - (5x - 1)(z + 2)}{(z + 2)(x + 2)} \right] \\
 &= \lim_{z \rightarrow x} \frac{1}{z - x} \left[\frac{(5xz - x + 10z - 2) - (5xz + 10x - z - 2)}{(z + 2)(x + 2)} \right] \\
 &= \lim_{z \rightarrow x} \frac{1}{z - x} \left[\frac{5xz - x + 10z - 2 - 5xz - 10x + z + 2}{(z + 2)(x + 2)} \right]
 \end{aligned}$$

SOLUTION:

$$\begin{aligned}
 & \bullet \\
 &= \lim_{z \rightarrow x} \frac{1}{z - x} \left[\frac{11z - 11x}{(z + 2)(x + 2)} \right] \\
 &= \lim_{z \rightarrow x} \frac{1}{z - x} \left[\frac{11(z - x)}{(z + 2)(x + 2)} \right] \\
 &= \lim_{z \rightarrow x} \frac{11}{(z + 2)(x + 2)} \\
 &= \frac{11}{(x + 2)^2}
 \end{aligned}$$

SOLUTION:

- At $x = 9$,

$$m = \frac{11}{(9 + 2)^2} = \frac{1}{11}$$

- Also, the point of tangency is $[9, f(9)]$. By substitution, $f(9) = \frac{5(9)-1}{(9)+2} = \frac{44}{11} = 4$. Thus, the point of tangency is $(9, 4)$. The equation of the line may be determined by using the point-slope equation:

SOLUTION:

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$$y - y_1 = m(x - x_1)$$

$$y - 4 = \frac{1}{11}(x - 9)$$

$$y = \frac{1}{11}x - \frac{9}{11} + 4$$

$$y = \frac{1}{11}x + \frac{35}{11}$$